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CH.SC.U4CSE25275

Question_5:

DS	Session	Arrays	Question Information	Level 1	Challenge 25

Problem Description:
For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters.

The English spellings for the numbers 0 to 12 are:
ZERO ONE TWO THREE FOUR FIVE SIX
SEVEN EIGHT NINE TEN ELEVEN TWELVE

Input Format:
Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line.

Output Format:
Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2 3 0999
```

EXPECTED OUTPUT

```
2 3 0999. E E H O R T T W
```

Test Case 2

INPUT (STDIN)

```
8 14 0999
```

EXPECTED OUTPUT

```
8 14 0999. E G H I T
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
char nums[13][256]
```

Test Case 2

KEYWORD

```
for(n=0;n<26;n++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC

40

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Code:

```
/*
Question_5:
Given jersey numbers, print the numbers followed by all distinct
letters needed to spell those numbers, in alphabetical order.
999 marks the end of input.
*/

#include <iostream>
#include <cstring>
using namespace std;

int main()
{
    char nums[13][256] = {
                            "ZERO", "ONE", "TWO", "THREE",
                            "FOUR", "FIVE", "SIX", "SEVEN",
                            "EIGHT", "NINE", "TEN", "ELEVEN", "TWELVE"
                        };

        int input[100];
        int count = 0;
        int x;

        while (cin >> x && x != 999)
        {
            input[count++] = x;
        }
        bool used[26] = {false};
        for (int i = 0; i < count; i++)
        {
            int num = input[i];
```

```cpp
            if (num >= 0 && num <= 12)
            {
                for (int j = 0; nums[num][j] != '\0'; j++)
                {
                    used[nums[num][j] - 'A'] = true;
                }
            }
            else if (num == 14)
            {
                string word = "FOURTEEN";
                for (char ch : word)
                {
                    used[ch - 'A'] = true;
                }
            }
        }
        for (int i = 0; i < count; i++)
        {
            cout << input[i] << " ";
        }
        cout << "0999. ";
        for(int n=0;n<26;n++)
        {
            if (used[n])
            {
                cout << char('A' + n) << " ";
            }
        }
        return 0;
}
```

Output:

Test Case_1:

```
2 3 0999
2 3 0999. E H O R T W
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
8 14 0999
8 14 0999. E F G H I N O R T U
PS C:\Users\Prabin Yadav\Documen
```